

# Badge exercise



A positive number  $n$  is called **practical** if any number  $k$  with  $1 \leq k < n$  can be obtained as sums of distinct divisors of  $n$

the proper divisors of 8 are  $\{1, 2, 4\}$

$$\begin{array}{rcl} 1 & = & 1 \\ 2 & = & 2 \\ 3 & = & 1 + 2 \\ 4 & = & 4 \\ 5 & = & 1 + 4 \\ 6 & = & 2 + 4 \\ 7 & = & 1 + 2 + 4 \end{array}$$

For example, the first practical numbers are: 1, 2, 4, 6, 8, 12, 16

1. Prove that 1 is the only odd practical number
2. Generate the list of all practical numbers in Haskell

*Hint:* a *powerset* function can be helpful